

MODEL COMPARISON REPORT

Project : Advanced Regression Model Comparison & Optimization Platform

1. Project Overview

Multiple regression algorithms were trained and evaluated for predicting residential house prices. Every model was trained using the same processed dataset to ensure a fair comparison.

2. Models Compared

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

3. Performance Comparison

Model	MAE	RMSE	R2 Score
Linear Regression	28000	38000	0.72
Decision Tree	32000	42000	0.69
Random Forest	18000	24000	0.79
Gradient Boosting	19500	25000	0.77
XGBoost	18800	24500	0.78
LightGBM	19000	24800	0.78
CatBoost	19200	24600	0.78

4. Model Analysis

Linear Regression

Simple baseline model. Performs well only when relationships are mostly linear.

Decision Tree

Easy to interpret but susceptible to overfitting on training data.

Random Forest

Highest R^2 score. Robust against overfitting and selected as the deployment model.

Gradient Boosting

Strong predictive capability but requires careful tuning.

XGBoost

Highly optimized boosting algorithm with excellent predictive performance.

LightGBM

Fast gradient boosting framework suitable for large datasets.

CatBoost

Excellent handling of categorical data with competitive performance.

5. Model Ranking

1. Random Forest
2. XGBoost
3. CatBoost
4. LightGBM
5. Gradient Boosting
6. Linear Regression
7. Decision Tree

6. Final Selection

Random Forest Regressor achieved the best balance between accuracy and stability. It produced the highest R^2 score while maintaining comparatively lower prediction errors. Therefore, Random Forest was selected as the final model for deployment in the Flask web application.

7. Business Recommendation

- Use Random Forest for production deployment.
- Continue monitoring prediction accuracy using new property data.
- Periodically retrain the model as market conditions change.
- Incorporate additional features such as location indices and economic indicators to improve future performance.

Conclusion

The comparative analysis demonstrates that ensemble-based regression algorithms outperform traditional linear methods for this dataset. Random Forest provided the most reliable and accurate predictions, making it the preferred model for practical house price estimation.